Niklas Koeppe

Integrated MS/PhD student in Brain and Cognitive Engineering, KAIST

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Research Focus

I study how reinforcement learning agents can continually acquire, retain, and revise useful behavior as their environments change. I seek architectural mechanisms that support long-term adaptation without steadily increasing computational load or requiring strong algorithmic priors, such as known task boundaries. I focus on the problem of plasticity loss, where networks lose the ability to learn a new task over time. My general research interests are continual learning, reinforcement learning, and evolving agents.

Education

KAIST, Daejeon, South Korea 2023–present

Integrated MS/PhD in Brain and Cognitive Engineering. Advisor: Prof. Sang Wan Lee. Expected: Feb 2028.

Radboud University, Nijmegen, Netherlands 2019–2022

BSc in Artificial Intelligence, *cum laude* (GPA: 8.24/10). Thesis: *A biologically-inspired recurrent neural network model for multi-source sound categorization and localization.*

Exchange semester (2022): Sungkyunkwan University, South Korea, in Applied Artificial Intelligence.

Publication

N. Koeppe, L. F. Vecchietti, D. Han, D. Li, and S. W. Lee. "Mitigating Plasticity Loss through Architectural Design in Continual Learning." *International Conference on Machine Learning (ICML)*, 2026.

Paper | Code

Research Experience

Microsoft Research Asia, Shanghai, China Mar–Jun 2026

Research Intern | Mentor: Dongqi Han. Architectural methods for preserving plasticity in continual RL.

Posters and Presentations

Interpolation Layer: An Architectural Solution to Plasticity Loss in Continual RL 2025

Poster, Korea Day 2025, Microsoft Research Asia. With L. F. Vecchietti and S. W. Lee.

How Dynamic Time Perception and Temporal Skipping Improve Efficiency in Reinforcement Learning 2024

Poster, AP-CCN 2024 and Korea Day 2024, Microsoft Research Asia. With S. W. Lee.

Teaching and Academic Service

Teaching Assistant, KAIST (Prof. Sang Wan Lee) 2024–2025

How AI and the Brain Work (2024, 2025); *Neuroscience-inspired Artificial Intelligence* (2025).

Session Coordinator, ELLIS UnConference, Paris: *Simulated vs. Unsimulated World Models* Dec 2026

Reviewer, NeurIPS 2026

Additional Experience

GLS IT Services GmbH, Eschborn, Germany (remote) Aug 2022–Jan 2023

Business Analyst, Product Management Operations. Applied statistical and machine learning methods to logistics data, including anomaly detection and handwritten signature verification.

Skills

Programming: Python (PyTorch, scikit-learn, Keras), Java, Scala, Lua, SQL; AWS.
Languages: German (native), English (fluent), Korean (basic), Chinese (basic).